

Aidan Ream

✉ aidanream01@gmail.com

☎ 720-371-2120

in aidan-ream

🔗 acream01

Overview

Computer Graphics Programmer with experience in both high-level applications and low-level implementation. Focused on a blended approach in Animation Systems programming and implementation.

Projects

Monte Carlo Ray Tracer

- Made a C++ Path Tracer with the following features: Geometric primitive and triangle-mesh support, performance optimization structures, multi-threading, png output, Probability Distribution Functions, Atmospheric Persepective camera effects, and more.
- Tools Used: C++

Compute Shader Based Vulkan Cloth Simluation

- Created a real time simulation environment with the Vulkan graphics library that utilizes a spring based compute shader on a grid of connected points to simulate the movement and flow of cloth using the GPU for performance optimization.
- Tools Used: C++, Vulkan

Wizard-Cat Library Video Game Engine

- Collaborated on a 3D Roguelike adventure game created by building a custom game engine. My primary contributions were to implement a skeletal animation system, 3D modeling, animating the characters, and creating tools for debugging.
- Tools Used: C++, openGL, Blender

Dynamic Dithering Shader

- Created a Shader Algorithm to introduce “dithering” printing effect to a dynamic 3D Scene
- Tools Used: C++, openGL, GLSL, GLFW

Technology Demo for Expressive Technology Studios

- Built an immersive technology demo in Unity combining motion capture, blender animation, spatial audio.
- Tools Used: Unity, Blender, Rokoko studio, Davinci Resolve

Experience

Calpoly, Lab Manager at Expressive Technology Studios

- Provided software updates and equipment upkeep for lab equipment over the course of 3.5 years
- Ran 10+ events per year utilizing our spatial audio theater space, assisting visiting speakers with the use of the software

San Luis Obispo, CA
August 2022 – December 2025

ACM SIGGRAPH, Student Volunteer

- Provided stage assistance during four performances in the 'Spatial Storytelling' Event
- Assisted conference attendees with navigation and scheduling

Vancouver, BC - August 2025
Los Angeles, CA - July 2026

Skills

Languages: C++, C, Java, Python, GLSL, JavaScript

Graphics Libraries & Game Engines: openGL, Vulkan, Unity, Godot, Unreal Engine 5, Houdini, RenderDoc

Visual Design Software: Maya, Blender, Substance Painter, Photoshop, Adobe Illustrator, Krita

Soft skills: Team management, Group Communication, Agile/Scrum workflows

Education

Cal Poly, San Luis Obispo CA
Bachelor of Science in Liberal Arts and Engineering Studies

Graduated June 2026
Cumulative GPA 3.54

Computer Graphics concentration - Coursework focused on computer graphics projects

2D and 3D Animation concentration - Coursework focused on design principles and industry standard workflows/technologies

Computer Science Minor - Additional coursework taken exploring various computer science disciplines